

CHORDAL-DELETION AND JOINED-OBSTRUCTION BOUNDS FOR THE COMPLETION NUMBER OF A GRAPH

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STATUS AND SOURCE PROBLEM

Bakonyi–Constantinescu [1] define the *completion number* $\text{cn}(G)$ as the least integer k such that every partial positive Hermitian matrix with graph G has a Hermitian completion with at most k negative eigenvalues. They ask for a graph-theoretic way to determine $\text{cn}(G)$ and, more specifically, for characterizations of the graphs with prescribed completion number.

This packet gives graph-theoretic lower and upper bounds that coincide on a large exact class. It is a partial answer, not a full classification.

Theorem 1.5. *every partial positive matrix with a chordal graph admits a positive completion.*

In [3] it was also shown that given any nonchordal graph G there exists a partial positive matrix A with graph G such that A does not admit a positive completion.

It appears as a potentially interesting question to ask what is the minimum number of negative eigenvalues one can always obtain in case of a given nonchordal graph.

Definition. Let G be an undirected graph. The *completion number* of G is the smallest number k such that every partial positive matrix A with graph G has a completion M with $i_-(M) \leq k$.

The following is our proposed Research Problem.

Question: Given a graph G , how can one find its completion number?

DEFINITIONS

A partial Hermitian matrix with graph G has every diagonal entry specified and has the off-diagonal entry a_{uv} specified exactly when $uv \in E(G)$. It is *partial positive* if its principal matrix on every clique of G is positive semidefinite.

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Let $\text{cvd}(G)$ be the *chordal vertex-deletion number*:

$$\text{cvd}(G) = \min\{|S| : S \subseteq V(G), G - S \text{ is chordal}\}.$$

Define the *joined-obstruction packing number* $\text{jp}(G)$ to be the largest integer r for which there are pairwise disjoint vertex sets $V_1, \dots, V_r$ such that

- (1) every induced graph $G[V_i]$ is nonchordal; and
- (2) for $i \neq j$, every vertex of V_i is adjacent to every vertex of V_j .

Thus the induced graph on $V_1 \cup \dots \cup V_r$ is the complete join of the nonchordal induced graphs $G[V_i]$.

IDEA OF THE PROOF

For the upper bound, delete a smallest set S making the graph chordal and positively complete the principal block outside S . Restoring one deleted row and column can increase the negative index by at most one, by Cauchy interlacing.

For the lower bound, place on each nonchordal joined block a partial positive matrix having no positive semidefinite completion. Specify all cross-block entries to be zero. Those entries exist because the blocks are completely joined. Hence every completion contains a block-diagonal principal matrix with at least one negative eigenvalue in each diagonal block.

THE GRAPH-THEORETIC SANDWICH

Theorem 1. *For every finite undirected graph G ,*

$$\text{jp}(G) \leq \text{cn}(G) \leq \text{cvd}(G).$$

Consequently, if $\text{jp}(G) = \text{cvd}(G)$, then this common graph-theoretic value is the completion number of G .

Proof. Set $k = \text{cvd}(G)$ and choose $S \subseteq V(G)$ with $|S| = k$ such that $H = G - S$ is chordal. Let A be any partial positive matrix with graph G . Its restriction $A[H]$ is partial positive. By the positive semidefinite completion theorem for chordal graphs [2], it has a positive semidefinite completion B .

Use B for the principal block indexed by $V(G) \setminus S$, retain every specified entry of A involving a vertex of S , and assign arbitrary values (say zero) to all remaining unspecified entries. This gives a Hermitian completion M of A containing the positive semidefinite matrix B as a principal submatrix of codimension k . Repeated Cauchy interlacing yields

$$i_-(M) \leq i_-(B) + k = k.$$

Since A was arbitrary, $\text{cn}(G) \leq \text{cvd}(G)$.

For the other inequality, let $V_1, \dots, V_r$ witness $r = \text{jp}(G)$ and write $H_i = G[V_i]$. Because H_i is nonchordal, the converse part of the chordal completion theorem gives a partial positive matrix A_i with graph H_i that has no positive semidefinite completion [2, 1].

Construct a partial matrix A with graph G as follows. Its restriction to V_i is A_i . Put 1 on the diagonal at every vertex outside the union of the V_i , and put 0 in every other specified off-diagonal position. This includes every position joining V_i to V_j , because the two blocks are completely adjacent.

The matrix A is partial positive. Indeed, on a clique K its specified principal matrix is the direct sum of the positive semidefinite clique restrictions $A_i[K \cap V_i]$ and scalar 1 blocks for the remaining vertices.

Let M be any Hermitian completion of A . Its principal submatrix on $V_1 \cup \dots \cup V_r$ is

$$M[V_1 \cup \dots \cup V_r] = M_1 \oplus \dots \oplus M_r,$$

because all cross-block entries were specified as zero. Here each M_i is a completion of A_i , so $i_-(M_i) \geq 1$. Therefore

$$i_-(M) \geq i_-(M[V_1 \cup \dots \cup V_r]) = \sum_{i=1}^r i_-(M_i) \geq r.$$

Thus $\text{cn}(G) \geq \text{jp}(G)$. $\square$

EXACT CLASSES

Corollary 2. *If G is nonchordal and $G - v$ is chordal for some vertex v , then $\text{cn}(G) = 1$.*

Proof. Nonchordality gives $\text{jp}(G) \geq 1$, while the hypothesis gives $\text{cvd}(G) \leq 1$. Apply Theorem 1. $\square$

Every chordless cycle is covered by Corollary 2. The source's graph on $4r$ vertices whose complement is a matching of size $2r$ is the join of r chordless four-cycles. Those factors witness $\text{jp}(G) \geq r$, so Theorem 1 recovers the sharp lower bound in the source's exact calculation $\text{cn}(G) = r$.

LIMITATIONS AND UPGRADE BARRIER

The bounds need not be equal. A tempting strengthening would claim that a join of chordal graphs is chordal, but this is false: the join of two independent two-vertex graphs is $K_{2,2}$. The checker caught the analogous failure for joins of cycles before promotion.

For example, if $m, n \geq 2$, then $K_{m,n}$ is nonchordal and

$$1 \leq \text{cn}(K_{m,n}) \leq \min(m-1, n-1).$$

The upper value is its chordal vertex-deletion number, while the joined-obstruction construction gives only one block. Ordinary vertex-disjoint induced cycles do not repair the lower bound: entries between such cycles may be unspecified, allowing a completion to couple their negative directions. This coupling phenomenon is the remaining obstruction to a full graph-theoretic characterization.

VERIFICATION AND NOVELTY NOTES

The proof was audited separately at the three sensitive steps: extension of a partial positive matrix after completing the chordal block, preservation of partial positivity in the joined-obstruction construction, and additivity of negative inertia on the fully specified zero cross blocks. A small brute-force graph checker verifies chordal-deletion number one for cycles and confirms that the join of two four-cycles has deletion number three, thereby falsifying the discarded join shortcut. The proof does not depend on it.

The local run indexes and arXiv source corpus had no duplicate. On 2026-08-11 an exact OpenAlex record lookup reported no citations to the source, and bounded exact-phrase searches found no later solution; searches were noisy because unrelated graph invariants share the phrase “completion number.” Novelty confidence is moderate.

REFERENCES

- [1] M. Bakonyi and T. Constantinescu, *Research problem: The completion number of a graph*, arXiv:math/0312390 (2003).
- [2] R. Grone, C. R. Johnson, E. M. Sá, and H. Wolkowicz, *Positive definite completions of partial Hermitian matrices*, Linear Algebra Appl. 58 (1984), 109–124.